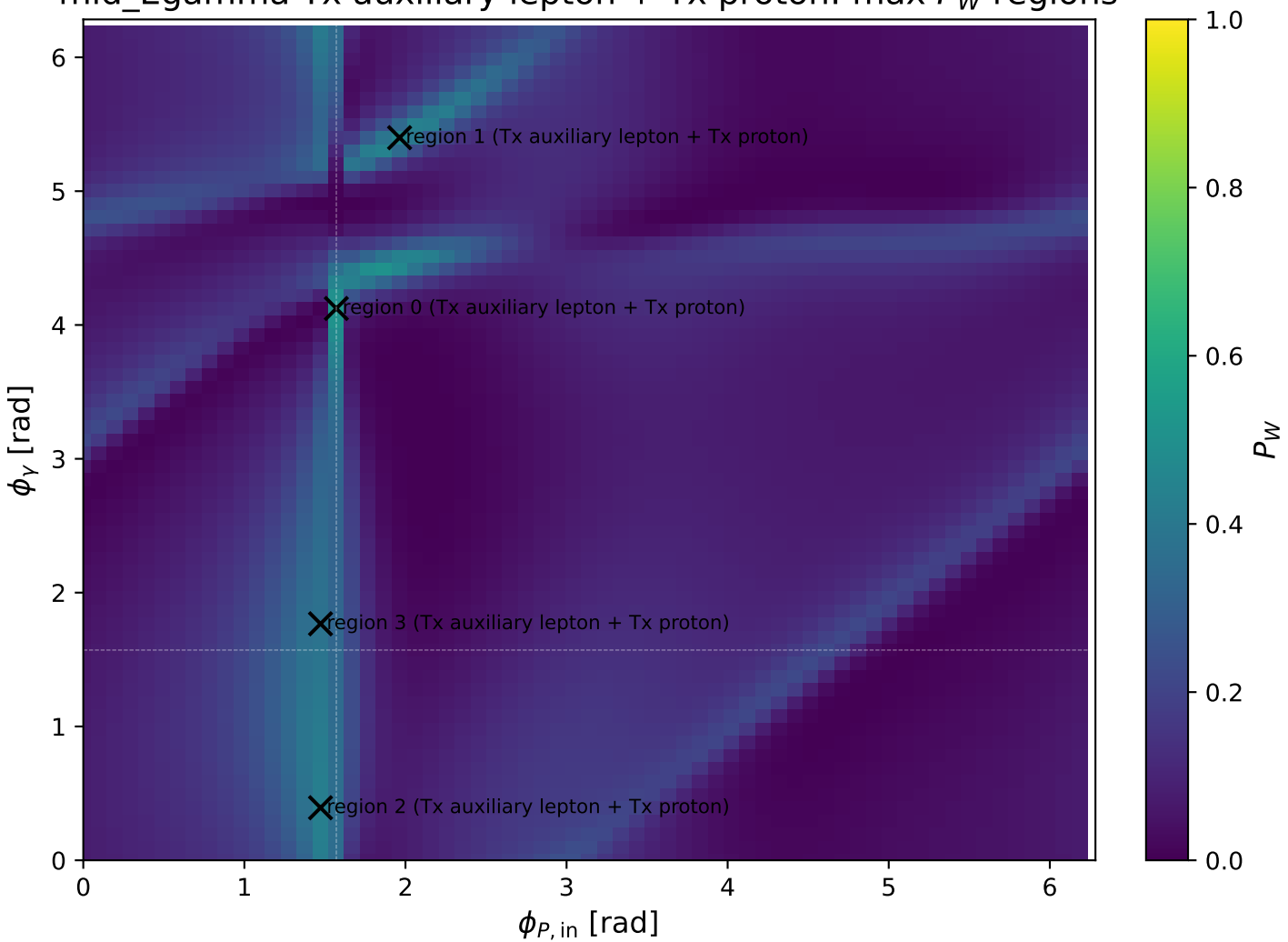
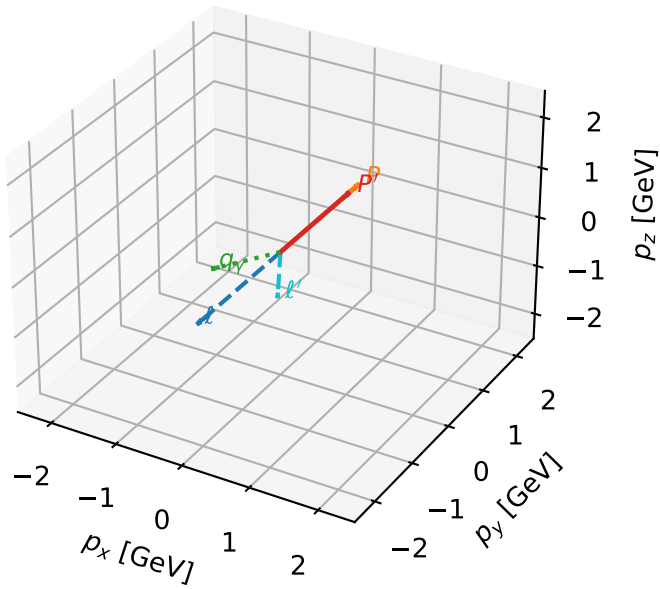


mid_Egamma Tx auxiliary lepton + Tx proton: max P_W regions



Tx auxiliary lepton + Tx proton characteristic max P_W configuration

3D momenta



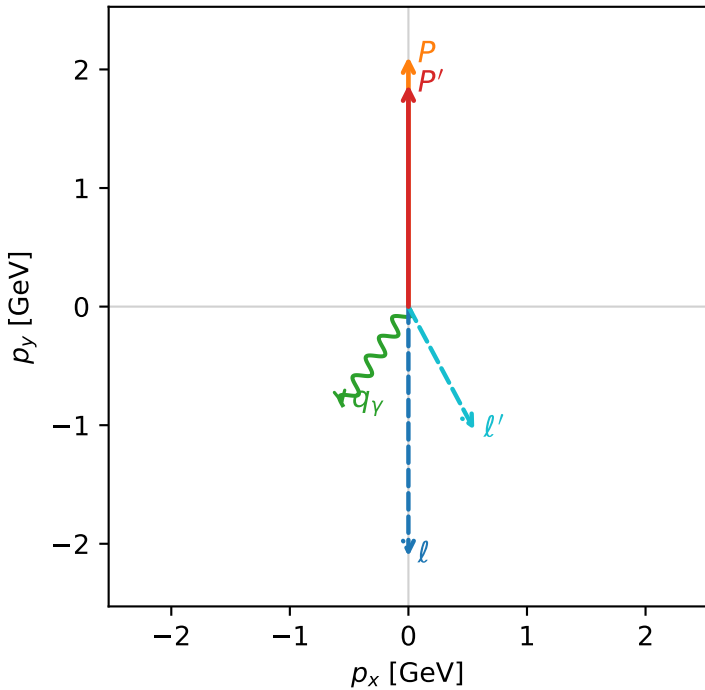
w_purity_mid_egamma_txtx_region_0 P_W=0.514377
 region: mid_Egamma_TxTx_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.87089$
 $\theta_{in}=1.57079$, $\phi_\ell=4.71239$, $\phi_p=1.5708$
 $E_\gamma=1$, $\phi_\gamma=4.12334$
 $Q^2=0.829595$, $x_B=0.102085$, $t=-0.0101624$, $W^2=8.17674$, $y=0.413857$
 $\theta(\ell', \gamma)=61.8745$ deg

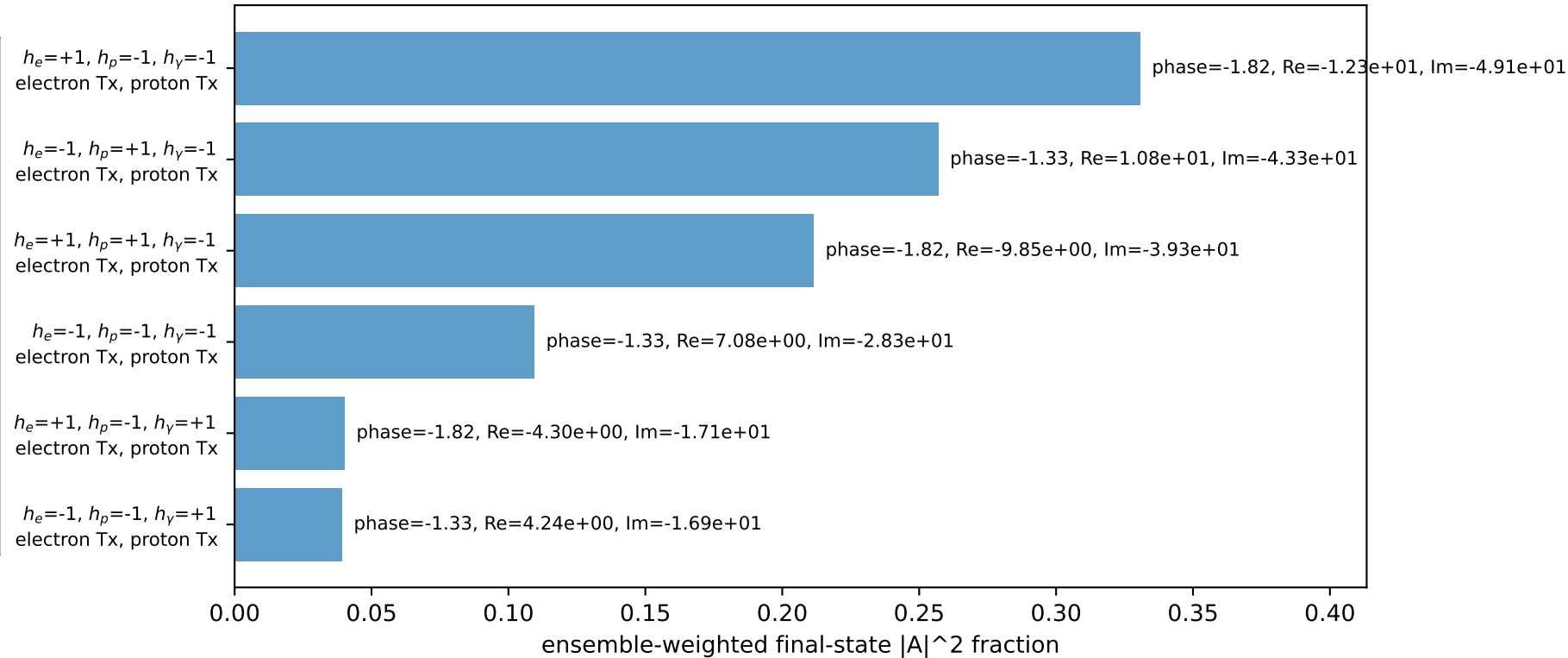
four-momenta $[E, p_x, p_y, p_z]$ GeV:

ℓ $E= 2.3322$ $p_x= -0.0000$ $p_y= -2.1069$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.0000$ $p_y= 2.1069$ $p_z= 0.0000$
 ℓ' $E= 1.5457$ $p_x= 0.5556$ $p_y= -1.0394$ $p_z= 0.0000$
 P' $E= 2.0929$ $p_x= 0.0000$ $p_y= 1.8709$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.5556$ $p_y= -0.8315$ $p_z= 0.0000$

Transverse plane

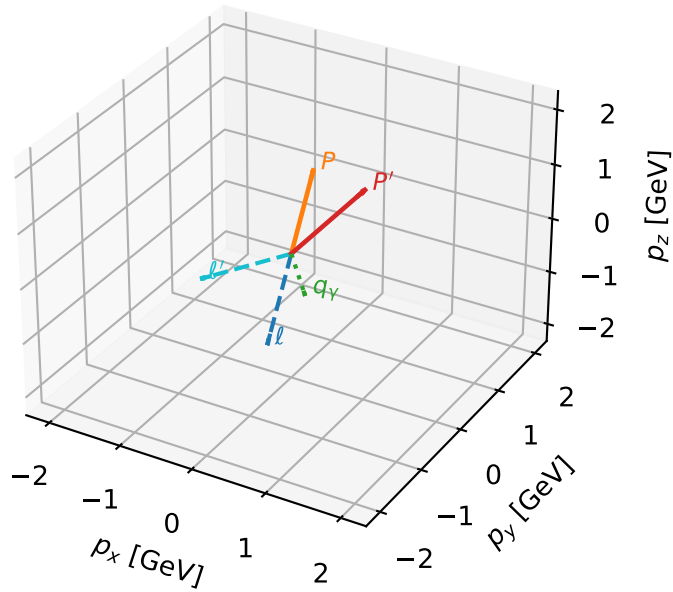


Leading initial-ensemble/final-state components (N=6, retained=98.8%)



Tx auxiliary lepton + Tx proton characteristic max P_W configuration

3D momenta

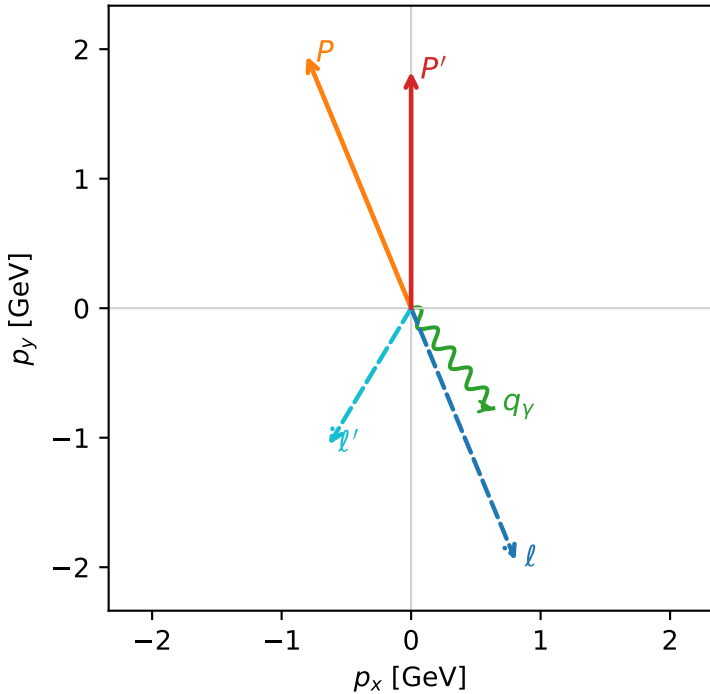


w_purity_mid_egamma_txtx_region_1 P_W=0.455715
 region: mid_Egamma_TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |T_{x,p}\rangle$

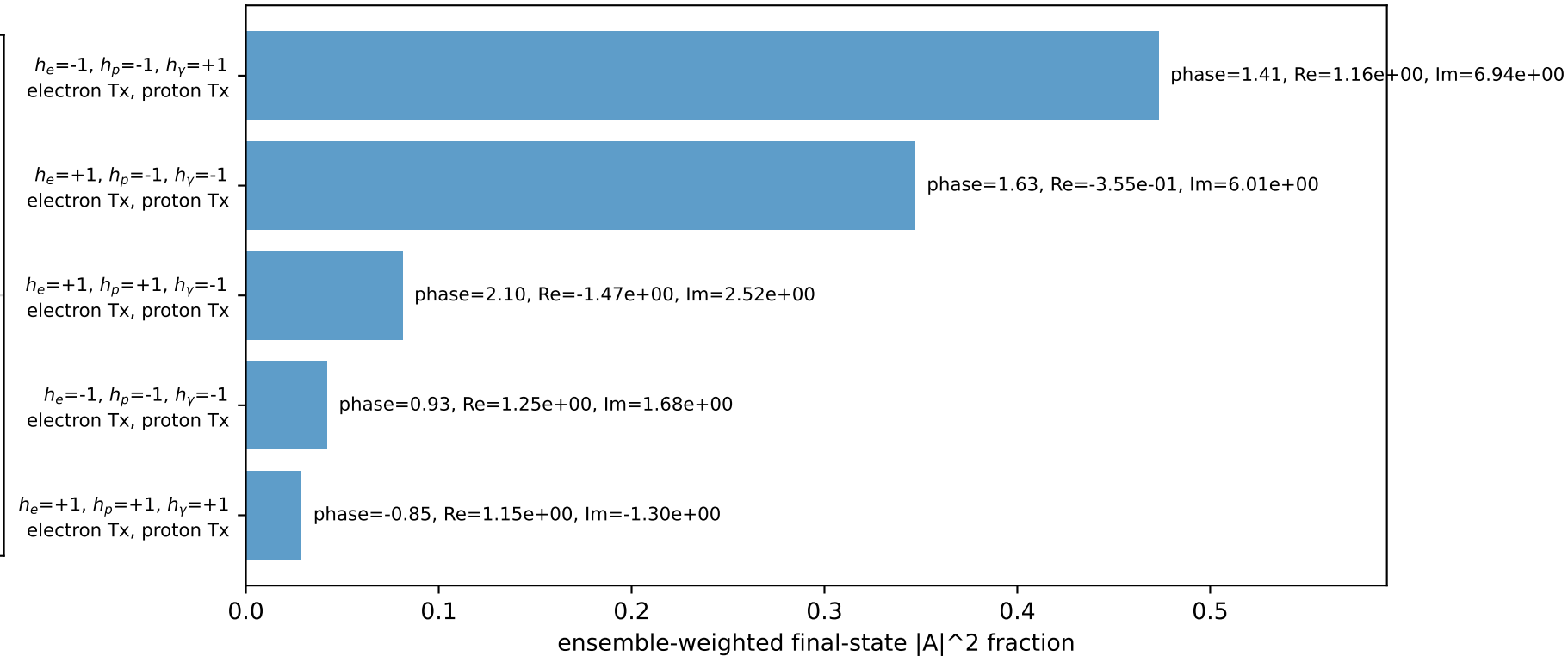
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.82664$
 $\theta_{in}=1.57079$, $\phi_i=5.10509$, $\phi_p=1.9635$
 $E_\gamma=1$, $\phi_\gamma=5.39961$
 $Q^2=2.31475$, $x_B=0.250363$, $t=-0.600527$, $W^2=7.81067$, $y=0.470848$
 $\theta(\ell', \gamma)=70.4273$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= 0.8063$ $p_y= -1.9466$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= -0.8063$ $p_y= 1.9466$ $p_z= 0.0000$
 ℓ' $E= 1.5851$ $p_x= -0.6344$ $p_y= -1.0536$ $p_z= 0.0000$
 P' $E= 2.0534$ $p_x= 0.0000$ $p_y= 1.8266$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.6344$ $p_y= -0.7730$ $p_z= 0.0000$

Transverse plane

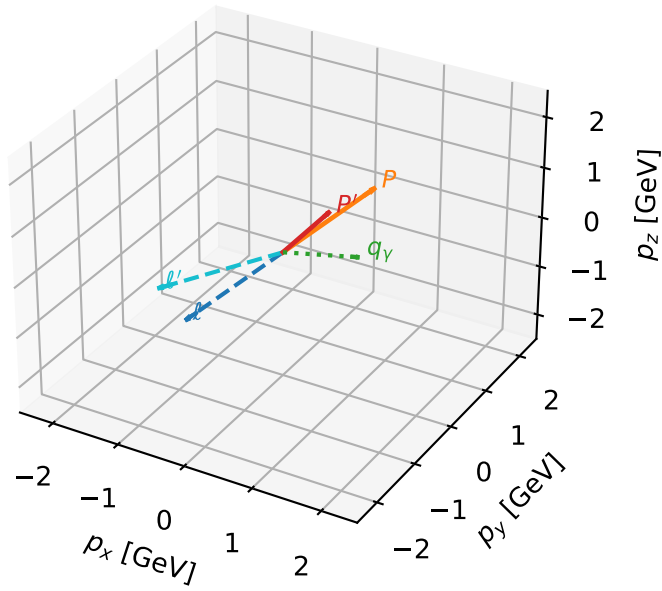


Leading initial-ensemble/final-state components (N=5, retained=97.3%)



Tx auxiliary lepton + Tx proton characteristic max P_W configuration

3D momenta

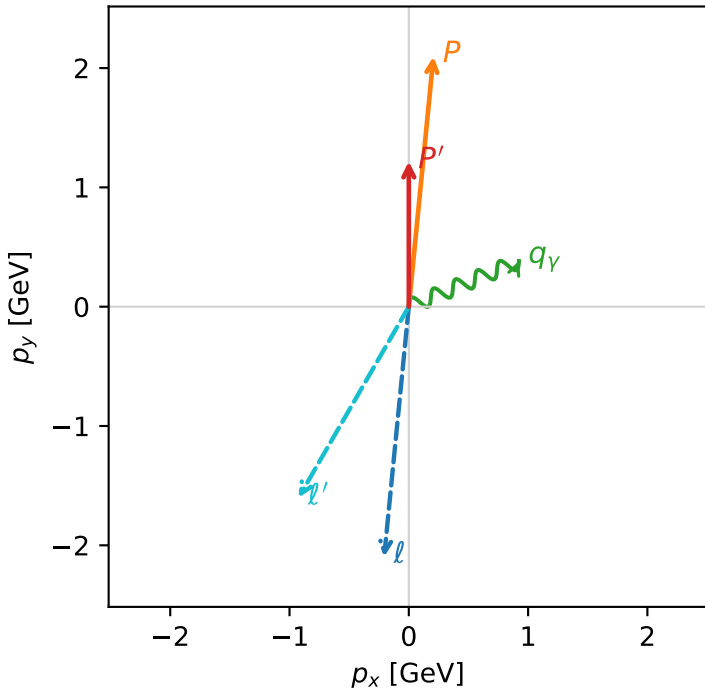


w_purity_mid_egamma_txtx_region_2 $P_W=0.426703$
 region: mid_Egamma_TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,l}\rangle \otimes |\bar{T}_{x,p}\rangle$

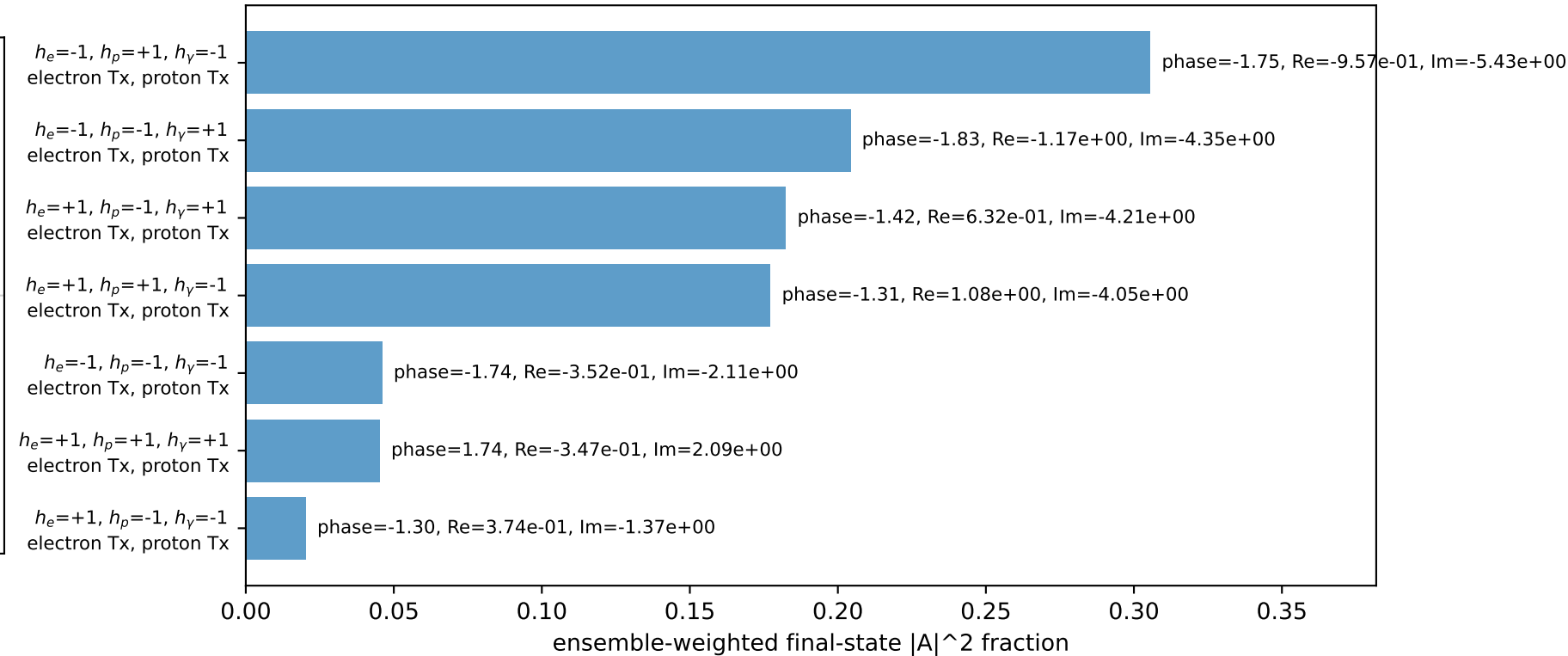
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.21789$
 $\theta_{in}=1.57079$, $\phi_l=4.61421$, $\phi_P=1.47262$
 $E_\gamma=1$, $\phi_\gamma=0.392699$
 $Q^2=0.707521$, $x_B=0.248265$, $t=-0.223661$, $W^2=3.02219$, $y=0.145135$
 $\theta(l', \gamma)=142.494$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.3322$ $p_x= -0.2065$ $p_y= -2.0968$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.2065$ $p_y= 2.0968$ $p_z= 0.0000$
 l' $E= 2.1013$ $p_x= -0.9239$ $p_y= -1.6006$ $p_z= 0.0000$
 P' $E= 1.5372$ $p_x= 0.0000$ $p_y= 1.2179$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.9239$ $p_y= 0.3827$ $p_z= 0.0000$

Transverse plane

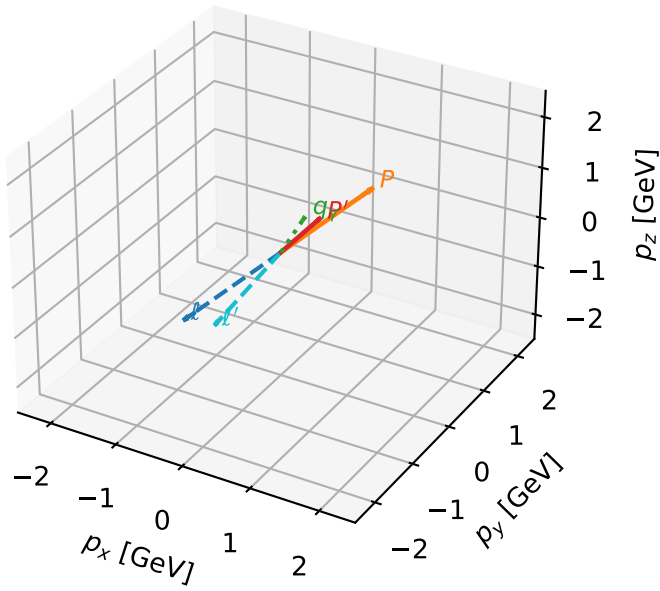


Leading initial-ensemble/final-state components (N=7, retained=98.1%)



Tx auxiliary lepton + Tx proton characteristic max P_W configuration

3D momenta

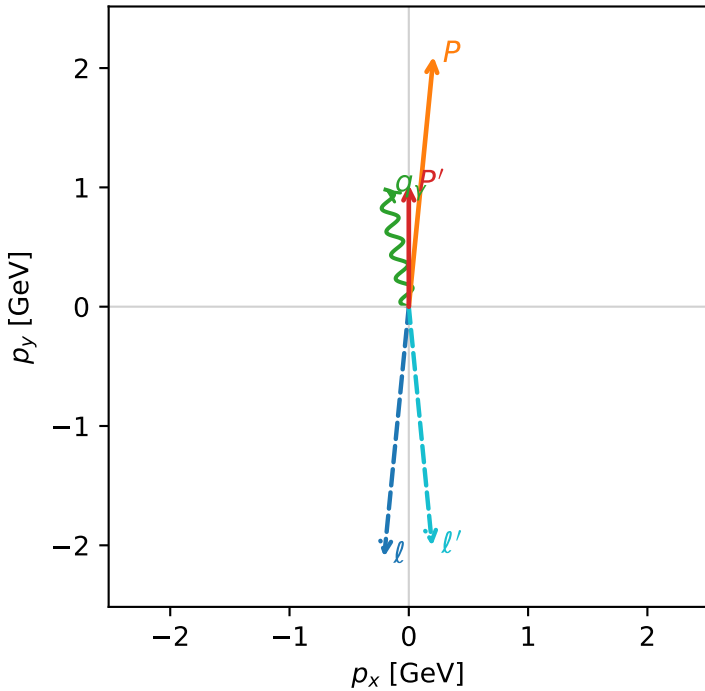


w_purity_mid_egamma_tctx_region_3 P_W=0.377877
 region: mid_Egamma_TxTx_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|T_{x,\ell}\rangle \otimes |\bar{T}_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.10694$, $|\vec{P}'|=1.02462$
 $\theta_{in}=1.57079$, $\phi_\ell=4.61421$, $\phi_P=1.47262$
 $E_Y=1$, $\phi_Y=1.76715$
 $Q^2=0.162779$, $x_B=0.17481$, $t=-0.351003$, $W^2=1.64824$, $y=0.0474221$
 $\theta(\ell', \gamma)=174.306$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.3322$ $p_x= -0.2065$ $p_y= -2.0968$ $p_z= -0.0000$
 P $E= 2.3063$ $p_x= 0.2065$ $p_y= 2.0968$ $p_z= 0.0000$
 ℓ' $E= 2.2494$ $p_x= 0.1951$ $p_y= -2.0054$ $p_z= 0.0000$
 P' $E= 1.3891$ $p_x= 0.0000$ $p_y= 1.0246$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= -0.1951$ $p_y= 0.9808$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_\gamma=-1$
 electron Tx, proton Tx

$h_e=-1, h_p=+1, h_\gamma=-1$
 electron Tx, proton Tx

$h_e=+1, h_p=-1, h_\gamma=+1$
 electron Tx, proton Tx

$h_e=-1, h_p=-1, h_\gamma=+1$
 electron Tx, proton Tx

$h_e=-1, h_p=+1, h_\gamma=+1$
 electron Tx, proton Tx

$h_e=+1, h_p=+1, h_\gamma=+1$
 electron Tx, proton Tx

$h_e=+1, h_p=-1, h_\gamma=-1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=7, retained=98.2%)

